

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Multivariable Calculus	Course Code:	MT2008
	Program:	Electrical Engineering	Semester:	Fall 2023
	Duration:		Total Marks:	
	Exam Date:	Sept 2023 (TBA by instructor)	Weight:	
	Section:	All	Page(s):	
	Exam Type:	Assignment quiz	CLO #	1

Student Name: _____ Roll No. _____ Section: _____

Instruction/Notes: 1. Do not forget to write your Name and Roll Numbers.
2. Solve on the paper and Return.

Problems from EX 12.5:

2, 5, 7, 12, 13, 18, 21, 22, 27, 30, 34, 36, 38, 39, 40, 42, 43, 55, 56, 57, 58, 65, 73, 74, 78

Question:

a) **Identify** (paraboloid, ellipsoid, etc.)

- | | |
|----------------------------|-------------------------------|
| 1. $x^2 + y^2 + 4z^2 = 10$ | 2. $z^2 + 4y^2 - 4x^2 = 4$ |
| 3. $9y^2 + z^2 = 16$ | 4. $y^2 + z^2 = x^2$ |
| 5. $x = y^2 - z^2$ | 6. $x = -y^2 - z^2$ |
| 7. $x^2 + 2z^2 = 8$ | 8. $z^2 + x^2 - y^2 = 1$ |
| 9. $x = z^2 - y^2$ | 10. $z = -4x^2 - y^2$ |
| 11. $x^2 + 4z^2 = y^2$ | 12. $9x^2 + 4y^2 + 2z^2 = 36$ |

b)

Drawing

Sketch the surfaces in Exercises

CYLINDERS

$x^2 + y^2 = 4$	$z = y^2 - 1$
$x^2 + 4z^2 = 16$	$4x^2 + y^2 = 36$

ELLIPSOIDS

$9x^2 + y^2 + z^2 = 9$	$4x^2 + 4y^2 + z^2 = 16$
$4x^2 + 9y^2 + 4z^2 = 36$	$9x^2 + 4y^2 + 36z^2 = 36$

PARABOLOIDS

$z = x^2 + 4y^2$	$z = 8 - x^2 - y^2$
$x = 4 - 4y^2 - z^2$	$y = 1 - x^2 - z^2$